

# HR Data Analytics Capstone

Executive Summary & Analysis | Author: Nalla Sumang

## 1. Business Objective

Salifort Motors is facing a critical employee retention issue. The Human Resources department requires data-driven interventions to identify flight-risk employees and the systemic causes of their departure. Hiring is expensive, and identifying the drivers of attrition is paramount for operational efficiency.

## 2. Data Diagnostics & Integrity

The raw dataset contained 14,999 employee records with 10 features (satisfaction levels, evaluations, project count, monthly hours, tenure, etc.). Rigorous data cleaning revealed severe data integrity issues:

- **Data Duplication:** A staggering 3,008 rows (20% of the dataset) were exact duplicates. These were dropped, as perfect replication across 10 distinct variables indicates a data collection error rather than genuine human survey responses.
- **Tenure Outliers:** 824 outlier records were detected in the tenure distribution, indicating cohorts that deviate significantly from standard employee lifecycles.

## 3. Actionable Insights & The "Burnout Metric"

Following data sanitization, the true baseline attrition rate stands at approximately 16.6% (1,991 departures). Exploratory Data Analysis (EDA) exposed the following trends:

**The 7-Project Death Spiral:** 100% of employees assigned to 7 projects left the company. Their average monthly hours far exceeded any other cohort (255-295 hours/month).

**The Bimodal Departure Distribution:** Departing employees fall into two distinct groups. Group A works abnormally low hours (likely fired or mentally checked out). Group B works exceptionally high hours, indicating severe burnout.

**The Sweet Spot:** Optimal retention occurs when employees are assigned to 3 or 4 projects.

## Unfiltered Analysis

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Systemic overwork is deeply embedded in the company culture. Assuming a standard 40-hour work week, an employee should log ~166.67 hours monthly. Excluding those on merely two projects, *all* cohorts—even those who stay—are clocking substantially more hours. The data indicates that Salifort Motors is over-leveraging its workforce. The 20% duplicate rate in raw HR data also suggests poor administrative tracking. Immediate product management and HR operational overhauls are required to address these foundational gaps before relying entirely on predictive models.